



Disorders of the Esophagus

❖ Gastro- oesophageal reflux disease (GERD)

- occurs when the amount of gastric juice that refluxes into the esophagus exceeds the normal limit, causing symptoms with or without associated esophageal mucosal injury.
- Gastro- oesophageal reflux resulting in heartburn affects approximately 15% of the general population.

Pathophysiology

- Occasional episodes of gastro- oesophageal reflux are common in healthy individuals.
- Reflux is normally followed by oesophageal peristaltic waves that efficiently clear the gullet, alkaline saliva neutralizes residual acid and symptoms do not occur .
- Gastro-oesophageal reflux disease develops when the esophageal mucosa is exposed to gastroduodenal contents for prolonged periods of time, resulting in symptoms and, in a proportion of cases, oesophagitis.
- Several factors are known to be involved in the development of gastro oesophageal reflux disease and these are shown next.

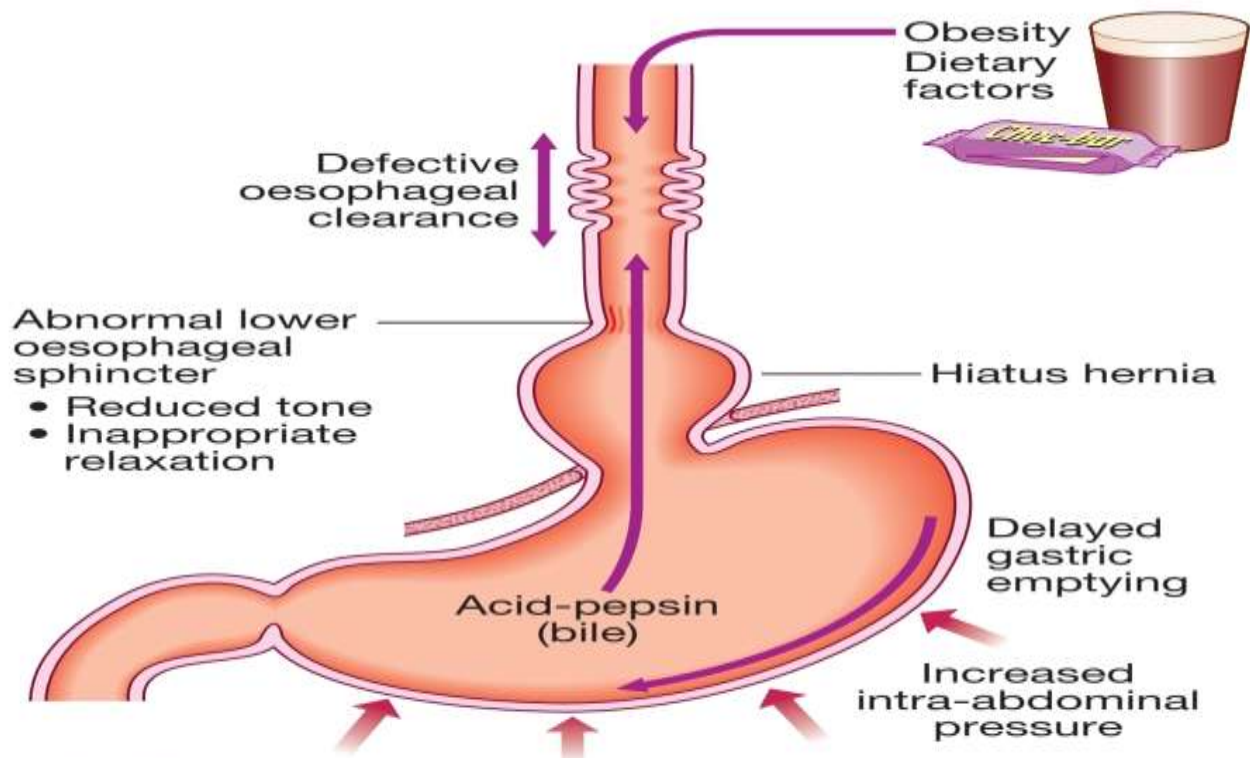


Fig. 21.26 Factors associated with the development of gastro-oesophageal reflux disease.



➤ **Abnormalities of the lower oesophageal sphincter**

The lower oesophageal sphincter is tonically contracted under normal circumstances, relaxing only during swallowing. Some patients with gastro oesophageal reflux disease have reduced lower oesophageal sphincter tone, permitting reflux when intra-abdominal pressure rises. In others, basal sphincter tone is normal but reflux occurs in response to frequent episodes of inappropriate sphincter relaxation.

➤ **Hiatus hernia**

Hiatus hernia causes reflux because the pressure gradient is lost between the abdominal and thoracic cavities, which normally pinches the hiatus. In addition , the oblique angle between the cardia and oesophagus disappears .

Many patients who have large hiatus hernias develop reflux symptoms but the relationship between the presence of a hernia and symptoms is poor.

Nevertheless ,almost all patients who develop oesophagitis, Barrett's oesophagus or peptic strictures are found to have a hiatus hernia.

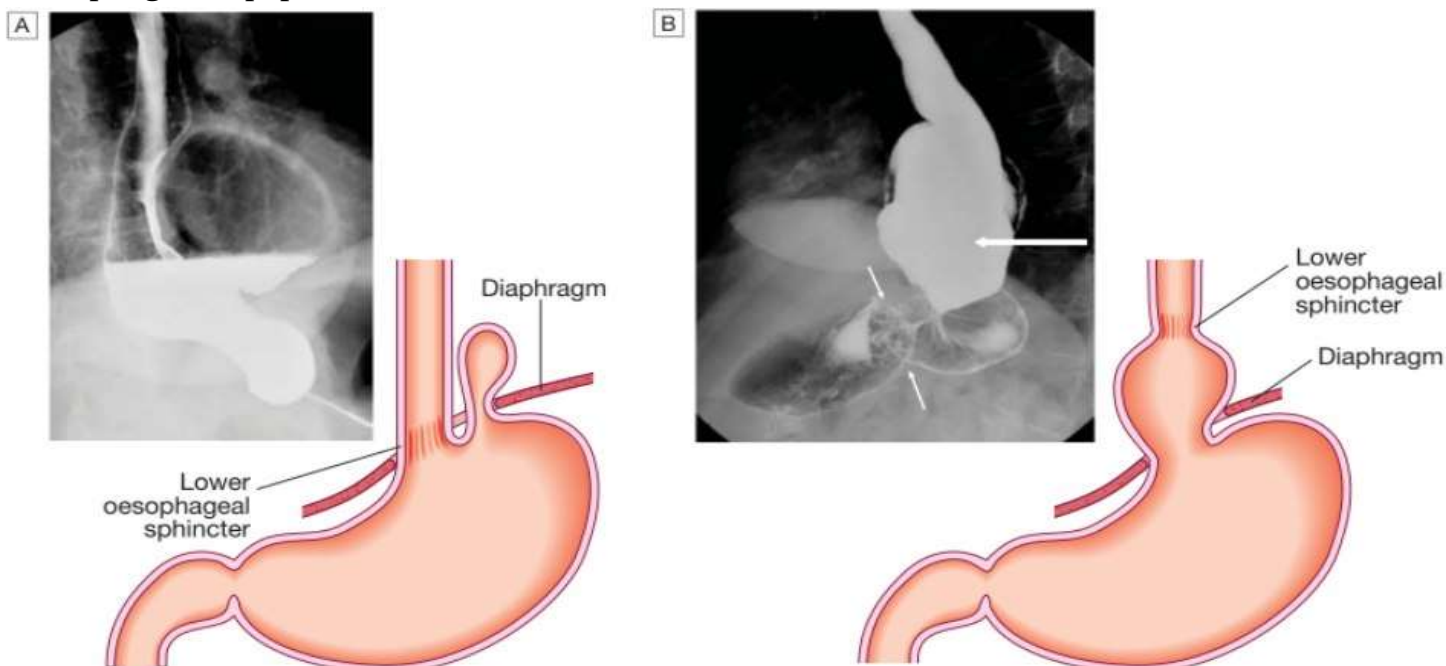


Fig. 21.27 Types of hiatus hernia. **A** Rolling or para-oesophageal. Inset: Barium meal showing a large para-oesophageal hernia with intrathoracic stomach. **B** Sliding. Inset: Barium meal showing a gastric volvulus (small arrows) complicating a sliding hiatus hernia (large arrow).

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21.29 Important features of hiatus hernia

- Herniation of the stomach through the diaphragm into the chest
- Occurs in 30% of the population over the age of 50 years
- Often asymptomatic
- Heartburn and regurgitation can occur
- Gastric volvulus may complicate large hernias



➤ Delayed oesophageal clearance

Defective oesophageal peristaltic activity is commonly found in patients who have oesophagitis. It is a primary abnormality, since it persists after oesophagitis has been healed by acid-suppressing drug therapy. Poor oesophageal clearance leads to increased acid exposure time.

➤ Gastric contents

Gastric acid is the most important oesophageal irritant and there is a close relationship between acid exposure time and symptoms. Pepsin and bile also contribute to mucosal injury.

➤ Defective gastric emptying

Gastric emptying is delayed in patients with gastro-oesophageal reflux disease. The reason is unknown.

➤ Increased intra-abdominal pressure

Pregnancy and obesity are established predisposing causes. Weight loss may improve symptoms.

➤ Dietary and environmental factors

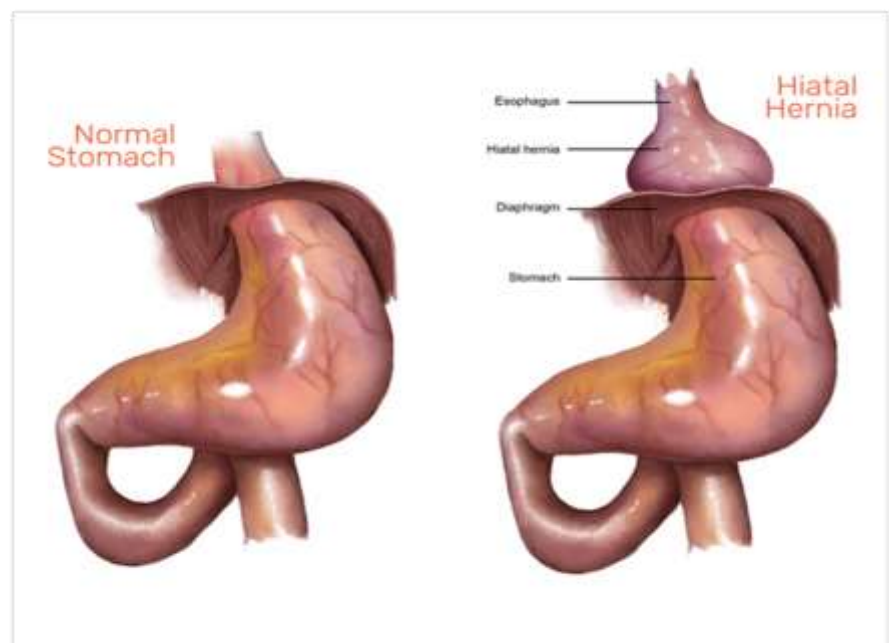
Dietary fat, chocolate, alcohol, tea and coffee relax the lower oesophageal sphincter and may provoke symptoms. The foods that trigger symptoms vary widely between affected individuals.

➤ Patient factors

Visceral sensitivity and patient vigilance play a role in determining symptom severity and consulting behaviour in individual patients.

❖ clinical features of GERD

- Heartburn&Regurgitation
- Water brush
- Chocking
- Dysphagia and odynophagia
- Chest pain
- Laryngitis
- Asthma!



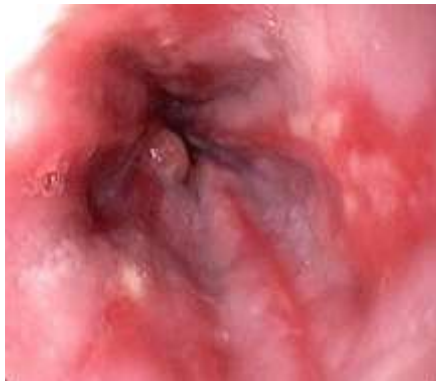


- Symptoms often provoked by bending, straining or lying down. 'Water brash', which is salivation due to reflux salivary gland stimulation as acid enters the gullet, is often present.
- GERD commoner in older people and in those who are overweight.
- The true relationship of these features to gastro-oesophageal reflux disease remains unclear.

❖ **Complication of GERD**

➤ **Oesophagitis**

A range of endoscopic findings is recognized, from mild redness to severe bleeding ulceration with stricture formation, although appearances may be completely normal. There is a poor correlation between symptoms and histological and endoscopic findings.



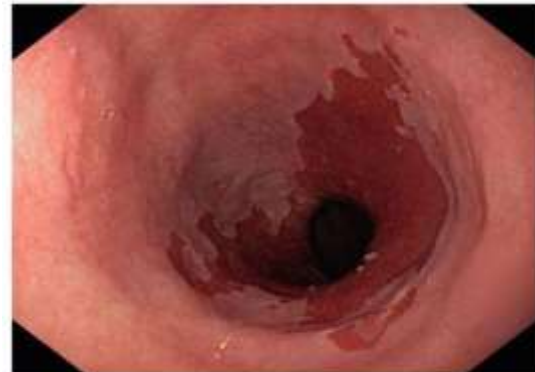
➤ **Barrett's oesophagus**

- Barrett's oesophagus is a pre-malignant condition, in which the normal squamous lining of the lower oesophagus is replaced by columnar mucosa (columnar lined oesophagus; CLO) that may contain areas of intestinal metaplasia. Patients are often asymptomatic until discovered, and can present with oesophageal cancer.
- Barrett's oesophagus is an adaptive response to chronic gastro oesophageal reflux and is found in 10% of patients undergoing gastroscopy for reflux symptoms.
- The estimated prevalence in Western countries is 1%–2% of the population.
- The relative risk of oesophageal cancer is increased 40–120-fold, but the absolute risk is low (0.1%–0.5% per year). The epidemiology and aetiology of Barrett's oesophagus are poorly understood.



Diagnosis:

Endoscopy is the gold standard for diagnosis. Multiple systematic biopsies are taken, in addition to sampling any visible lesions, to maximize the chance of detecting intestinal metaplasia and/or dysplasia.



Management:

- Neither potent acid suppression nor anti-reflux surgery stops progression or induces regression of Barrett's oesophagus, and treatment is indicated only for symptoms of reflux or complications, such as stricture.
- Regular endoscopic surveillance can detect dysplasia at an early stage and may improve survival but, because most Barrett's oesophagus is undetected until cancer develops, surveillance strategies are unlikely to influence the overall mortality rate of oesophageal cancer.
- Surveillance intervals for Barrett's oesophagus are risk stratified by the **length** of the Barrett's oesophagus segment, and by the presence and **grade of dysplasia**. For those with high-grade dysplasia or intramucosal carcinoma, the treatment options are either oesophagectomy or endoscopic therapy, with a combination of endoscopic resection of any visibly abnormal areas and radiofrequency ablation (RFA) of the remaining Barrett's mucosa, as an 'organ-preserving' alternative to surgery.



➤ Anemia

• Iron deficiency anemia can occur as a consequence of occult blood loss from long-standing oesophagitis. Most patients have a large hiatus hernia and bleeding can stem from subtle erosions in the neck of the sac ('Cameron lesions'). Nevertheless, hiatus hernia is very common and **other causes of blood loss, particularly colorectal cancer, must be considered in anemic patients**, even when endoscopy reveals oesophagitis.

➤ Benign oesophageal stricture

Fibrous strictures can develop as a consequence of long-standing oesophagitis.

The typical presentation is with dysphagia that is worse for solids than for liquids. A history of heartburn is common but not invariable; many patients presenting in old age with strictures have no preceding heartburn.

Diagnosis is by endoscopy, when biopsies of the stricture can be taken to exclude malignancy. Endoscopic balloon dilatation is helpful. Subsequently, long-term therapy with a PPI at full dose should be started to reduce the risk of recurrent oesophagitis and stricture formation.

i	21.31 Causes of oesophageal stricture
	• Gastro-oesophageal reflux disease • Webs and rings • Carcinoma of the oesophagus or cardia • Eosinophilic oesophagitis • Extrinsic compression from bronchial carcinoma • Corrosive ingestion • Post-operative scarring following oesophageal resection • Post-radiotherapy • Following long-term nasogastric intubation • Bisphosphonates

➤ Gastric volvulus

Occasionally, a massive intrathoracic hiatus hernia may twist on itself, leading to a gastric volvulus. This gives rise to complete oesophageal or gastric obstruction and the patient presents with severe chest pain, vomiting and dysphagia. The diagnosis is made by chest X-ray (air bubble in the chest) and barium swallow. A nasogastric tube is normally inserted in the acute phase to facilitate decompression, with surgery usually advised after the acute episode. Endoscopic decompression may be an alternative option in selected patients in old age.

❖ Investigations of GERD

Therapeutic trial

Endoscopy

Esophageal pH monitoring



- Young patients who present with typical symptoms of gastro oesophageal reflux, without worrying features such as **dysphagia**, **weight loss** or **anemia**, can be treated empirically without investigation. Investigation is advisable if patients present over the age of 50–55 years, if symptoms are **atypical** or if a **complication** is suspected.
- **Endoscopy** is the investigation of choice. This is performed to exclude other upper gastrointestinal diseases that can mimic gastro oesophageal reflux and to identify complications. A normal endoscopy in a patient with compatible symptoms should not preclude treatment for gastro-oesophageal reflux disease.
- Oesophageal pH monitoring is indicated if the diagnosis is unclear or surgical intervention is under consideration. This involves tethering a slim catheter with a terminal radiotelemetry pH-sensitive probe above the gastro-oesophageal junction; wireless oesophageal pH monitoring is available in some centers for those intolerant of catheter-based monitoring.
- Manometry helps surgical planning by determining the lower esophageal sphincter (LES) pressure and identifying any esophageal motility disorders.

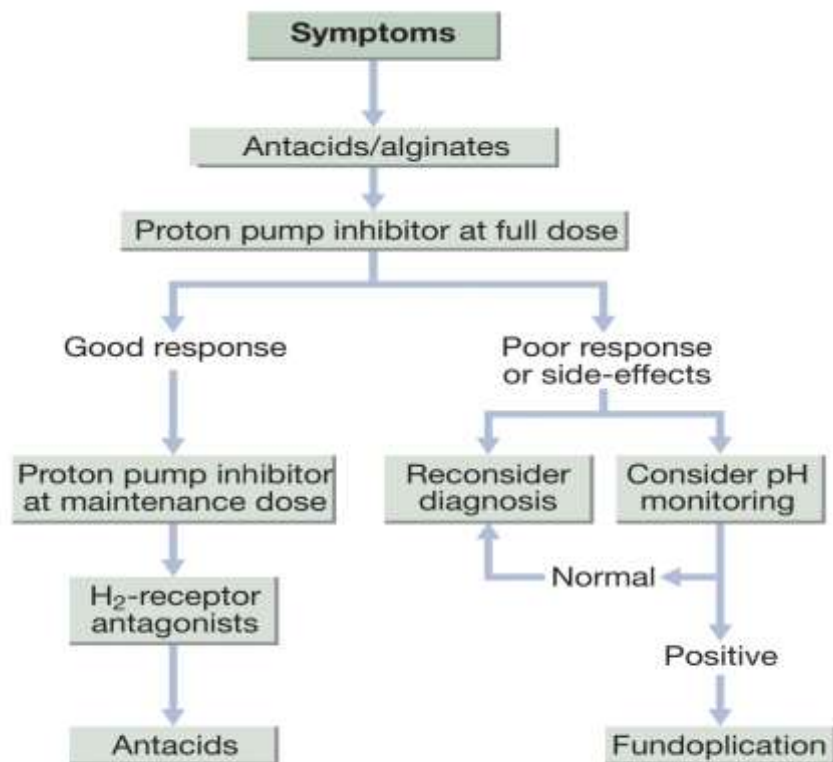
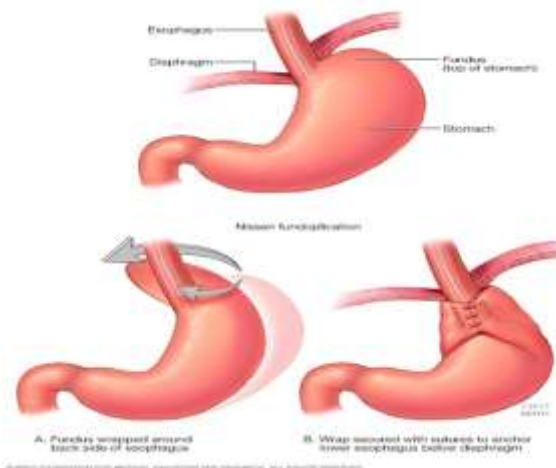
❖ **Management of GERD****Medical:****Surgical ; fundoplication:**

Fig. 21.30 Treatment of gastro-oesophageal reflux disease: a 'step-down' approach.

Lifestyle modifications

include the following:

- Losing weight (if overweight)
- Avoiding smoking, alcohol, chocolate, citrus juice, and tomato-based products (2005 guidelines from the American College of Gastroenterology [ACG] also suggest avoiding peppermint, coffee, and possibly the onion family)
- Avoiding large meals
- Waiting 3 hours after a meal before lying down
- Elevating the head of the bed



- Patients who fail to respond to these measures should be offered **PPIs**, which are usually effective in resolving symptoms and healing oesophagitis. Recurrence of symptoms is common when therapy is stopped and some patients require life-long treatment at the lowest acceptable dose.
- When dysmotility features are prominent, prokinetics such as **domperidone** or metoclopramide can be helpful. Proprietary antacids and alginates can also provide symptomatic benefit. H₂-receptor antagonist drugs relieve symptoms without healing oesophagitis.
- Long-term PPI therapy can lead to the development of parietal cell hyperplasia and hypertrophy, leading to **acid rebound** on withdrawal of PPIs after long-term usage.
- PPI therapy is also associated with **reduced absorption of iron, B12 and magnesium**. The drugs may also predispose to enteric infections with Salmonella, Campylobacter and possibly Clostridioides difficile.
- An association between **microscopic colitis** and PPIs has also been suggested, as well as with **acute interstitial nephritis**. Long-term therapy increases the risk of Helicobacter-associated progression of gastric mucosal atrophy and H. pylori eradication is advised in patients requiring PPIs for more than 1 year.
- Patients who fail to respond to medical therapy, those who are unwilling to take long-term PPIs and those whose major symptom is severe regurgitation should be considered for laparoscopic fundoplication.

Other causes of oesophagitis:

Infection: Oesophageal candidiasis occurs in debilitated patients and those taking broad spectrum antibiotics or cytotoxic drugs. It is a particular problem in patients with HIV/AIDS, who are also susceptible to a spectrum of other oesophageal infections.

Corrosives: Suicide attempt by ingestion of caustic acid (e.g. vinegar) or alkaline agents (e.g. bleach) is followed by painful burns of the mouth and pharynx and by extensive erosive oesophagitis. Late complications include oesophageal strictures, which may require endoscopic dilatation or stent insertion, or surgery in severe cases.



Drugs: Potassium supplements and NSAIDs may cause oesophageal ulcers when the tablets are trapped above an oesophageal stricture. Liquid preparations of these drugs should be used in such patients. Bisphosphonates cause oesophageal ulceration and should be used with caution in patients with known oesophageal disorders. Tetracyclines, such as doxycycline, can also cause oesophagitis.

Eosinophilic oesophagitis: This is an increasingly recognized condition, and is more common in males. It occurs more often in atopic individuals and is characterized by eosinophilic infiltration of the oesophageal mucosa. Patient may present with dysphagia or food bolus obstruction more often than heartburn. Endoscopy is usually normal, but **mucosal rings**, longitudinal furrows, exudates, oedema, strictures. To obtain a diagnosis, multiple biopsies should be taken from different levels of the oesophagus. Dietary modifications are an effective first-line treatment for this condition, and include elimination diets. Pharmacological management trial of **high-dose PPI** can be used in the first instance. In patients who do not respond, 8–12 weeks of therapy with **topical glucocorticoids** can be used delivered by spraying into the mouth via a metered-dose inhaler, and swallowed rather than inhaled.

Benign oesophageal stricture

- Benign oesophageal stricture is usually a consequence of gastroesophageal reflux disease and occurs most often in older adults who have poor oesophageal clearance. Rings, caused by submucosal fibrosis, are found at the oesophago-gastric junction ('Schatzki ring') and cause intermittent dysphagia, often starting in middle age.
- A post-cricoid web is a rare complication of iron deficiency anemia (Paterson–Kelly or Plummer–Vinson syndrome), and may be complicated by the development of squamous carcinoma.
- Benign strictures can be treated by endoscopic dilatation, in which wire-guided balloons are used to disrupt the fibrous tissue of the stricture.





❖ Esophageal motility disorders

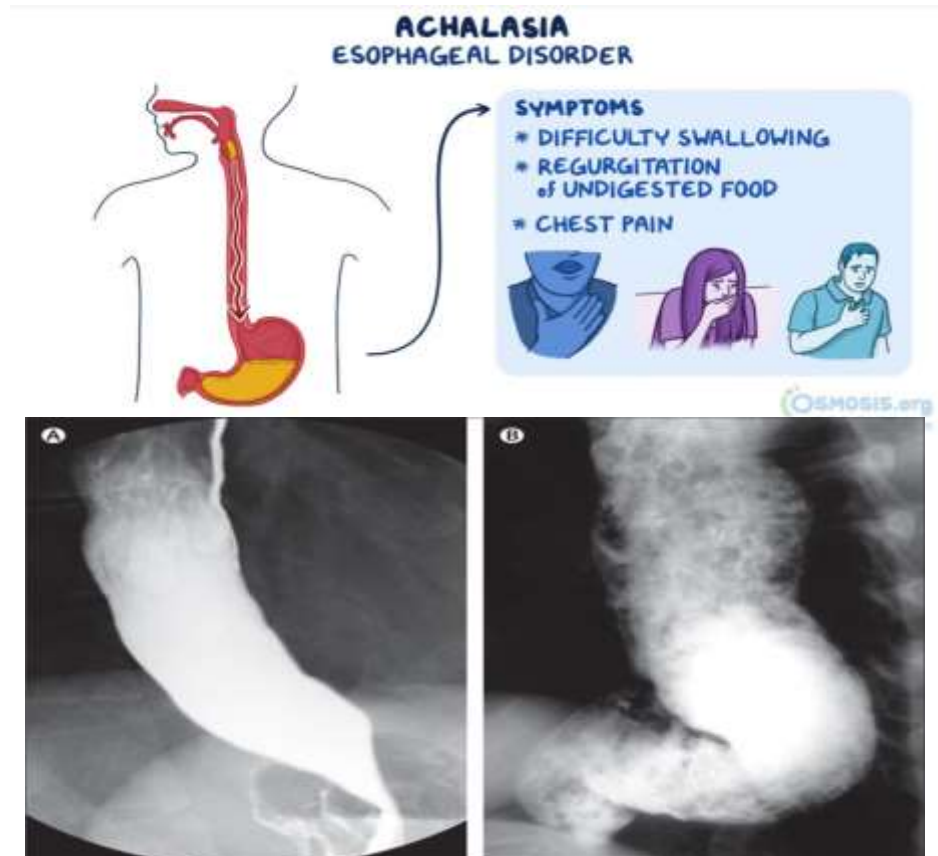
Achalasia of the oesophagus

Pathophysiology

- Achalasia is characterised by: a hypertonic lower oesophageal sphincter, which fails to relax in response to the swallowing wave, failure of propagated oesophageal contraction, leading to progressive dilatation of the gullet.
- The cause is unknown. It is associated with autoimmune diseases, such as type 1 diabetes mellitus, systemic lupus erythematosus, rheumatoid arthritis and Sjogren syndrome.
- Defective release of nitric oxide by inhibitory neurons in the lower oesophageal sphincter has been reported, and there is degeneration of ganglion cells within the sphincter and the body of the oesophagus.
- Loss of the dorsal vagal nuclei within the brainstem can be demonstrated in later stages.
- Infection with *Trypanosoma cruzi* in Chagas' disease causes a syndrome that is clinically indistinguishable from achalasia.

Clinical features

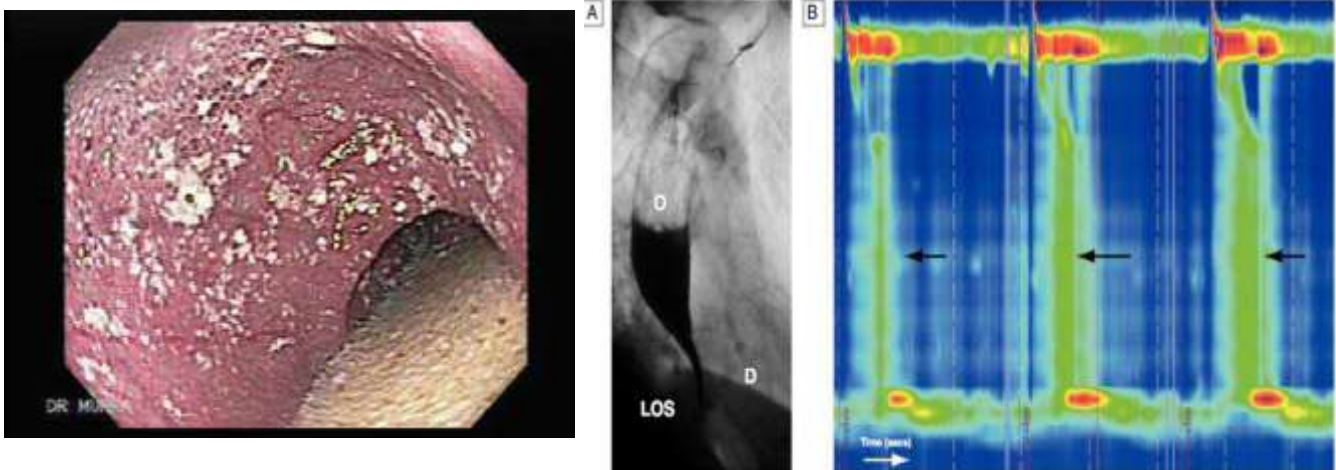
- Dysphagia
- Chest pain
- Wt loss
- pulm aspirate
- Squamous cell Ca





Investigations

- **Endoscopy** should always be carried out because carcinoma of the cardia can mimic the presentation and radiological and manometric features of achalasia ('pseudo-achalasia').
- A **barium swallow** shows tapered narrowing of the lower oesophagus and, in late disease, the oesophageal body is dilated, aperistaltic and food filled.
- **Manometry** confirms the high-pressure, non-relaxing lower oesophageal sphincter with poor contractility of the oesophageal body.



Management

Endoscopic : Forceful pneumatic dilatation using a balloon disrupts the oesophageal sphincter and improves symptoms in 80% of patients. Some patients require more than one dilatation, but those needing frequent dilatation are best treated surgically. Endoscopically directed injection of botulinum toxin into the lower oesophageal sphincter induces clinical remission, but relapse is common, with worsening response on repeated treatments. It tends to be reserved for frail individuals in old age where other treatments are too risky. Peroral endoscopic myotomy (POEM) is a newer advanced endoscopic technique for the management of achalasia, but is currently only performed in specialist centres.

Surgical : Surgical myotomy (Heller's operation), commonly performed laparoscopically or as an open operation, is effective but is more invasive than endoscopic dilatation. myotomy may be complicated by gastro oesophageal reflux, and this can lead to severe oesophagitis because oesophageal clearance is



so poor. For this reason, Heller's myotomy is accompanied by a partial fundoplication anti-reflux procedure. PPI therapy is often necessary after surgery.

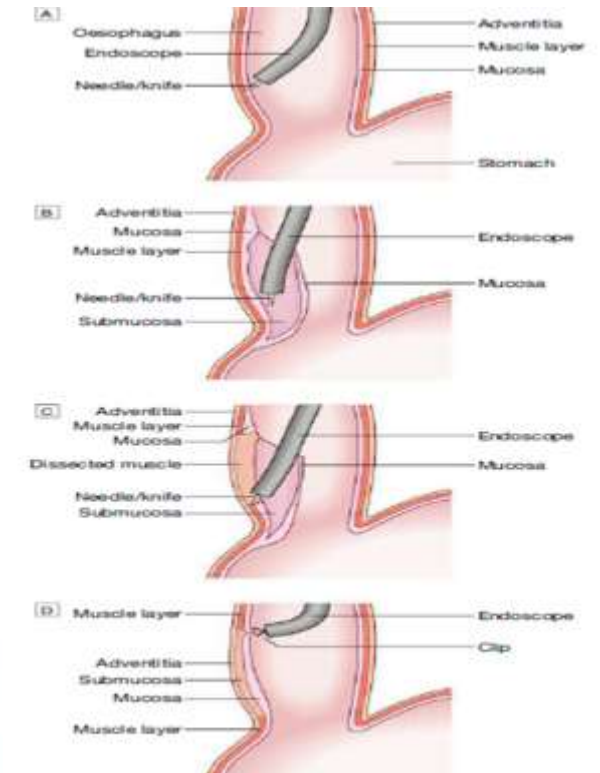
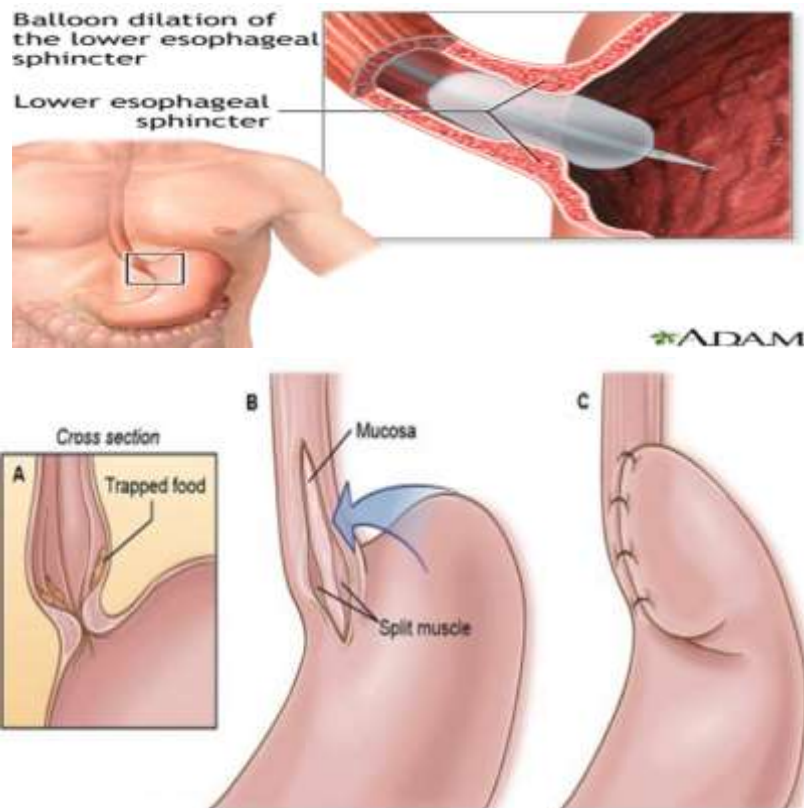


Fig. 23.35 Per-oral endoscopic myotomy. [A] Mucosal surface opened to access submucosal space. [B] A submucosal tunnel is created. [C] Circular muscle is dissected - endoscopic myotomy. [D] The entry site is closed with clips.

Pharyngeal pouch

This occurs because of incoordination of swallowing within the pharynx, which leads to herniation through the cricopharyngeus muscle and formation of a pouch. It is rare, affecting 1 in 100 000 people; it occurs predominantly in men, and is most common in the seventh and eighth decades of life. Many patients have no symptoms, but regurgitation, **halitosis** and **dysphagia** can be present. Some notice **gurgling** in the throat after swallowing. The investigation of choice is a **barium swallow** which demonstrates the pouch and reveals incoordination of swallowing, often with **pulmonary aspiration**. Treatment can be via a surgical approach, such as (diverticulotomy).

